

Circuit and working

Fig. 3 shows the circuit diagram of the pillbox with an adjustable reminder, designed to provide a daily audio alert in a sequential manner. It is built around two NE555 timers, a CD4017 decade counter, BC549 transistors, optocoupler PC817, a speaker, and a few other components.

The first NE555(A) is configured

in astable mode to generate a long-duration pulse of about 24 hours using a high-value RC network. This pulse serves as a daily clock signal for the CD4017 decade counter, which advances its outputs (Q0 to Q6) one step at a time, thereby selecting a different indicator for each day of the week and automatically resetting after the seventh output (Saturday).

The selected output is routed through an LED to the collector of the BC549 transistor. The output of NE555(A) directly triggers NE555(B) timer. This timer operates in monostable mode and produces a fixed pulse of around ten seconds, ensuring that the alert remains active only for a short duration. The monostable output is fed to an optocoupler (PC817), which provides electrical isolation between the low-power control section and the output stage, improving reliability and protecting the circuit from noise or transients.

The optocoupler drives the BT66 audio module, which plays a

pre-recorded message or tone. The audio output is further amplified by a BC549 transistor to drive the speaker, producing a clear audible alert. LEDs connected to the CD4017 outputs indicate the active day, making the system easy to monitor.

The circuit delivers a ten-second reminder once every 24 hours, progressing through different LEDs each day. Fig. 4 (timing diagram) shows that the system remains idle for most of the day and activates only during the short alert period, ensuring low power consumption and consistent operation.

Although the design is simple and cost-effective, the use of large-value resistors and electrolytic capacitors for 24-hour timing may introduce drift. For higher accuracy, a crystal-controlled or microcontroller-based timing approach may be preferable.

The complete pin-to-pin connections of the main components are given in Tables 2 and 3. The IC 555A operates in astable mode to generate a 24-hour clock pulse. Its pins 1, 4, and 8 are connected to ground and +5V as required, while pins 2 and 6 are tied together with a 4700 μ F capacitor to ground.

Pin 7 is connected to pin 6 through a 1k Ω resistor and to +5V via a 26.6M Ω resistor, and pin 5 is bypassed with a 0.01 μ F capacitor.

TABLE 1
BILL OF MATERIALS

Components	Quantity	Purpose
555 timer IC	2	One for astable (24 hours), one for monostable (10 sec)
CD4017 IC	1	For 7-day LED sequencing
BT66 music IC	1	Alert sound via the speaker
BC549 transistor	8	To switch the LEDs on/off and for the speaker
PC817 IC	1	To isolate BT66 from 555 IC
5mm LEDs	7	One/day: Monday to Sunday
Resistor R1 (26.6M Ω)	1	For 24-hour time in 555 IC
Resistor R2 (1K Ω)	1	For 24-hour time in 555 IC
Resistor (1M Ω)	1	For 10-sec time in 2nd 555 IC
Capacitor (4700 μ F)	1	For 555 timing (24-hour)
Capacitor (9/10 μ F)	1	For 555 timing (10-sec)
Capacitor (0.0 μ F)	2	For noise removal
Speaker (8 Ω , 0.5W)	1	For playing the sound
Breadboard	2	For prototyping the circuit
Jumper wires	-	For all connections
Power supply (5V)	1	Battery or DC adaptor
Diode 1N4007	1	To protect IC PC817

TABLE 2
MAIN COMPONENT PIN CONNECTIONS

Connection	IC555A (24-hour Astable)	IC555B (10-second Monostable)	CD4017	PC817	BT66	Speaker	BC549
GND	Pin 1	Pin 1	Pin 8	Pin 2	–	Negative (–)	Emitter
+5V	Pin 8, Pin 4	Pin 8, Pin 4	Pin 16	–	Pin 4	Positive (+)	–
Control Voltage	Pin 5 $\rightarrow$ 0.01 μ F to GND	Pin 5 $\rightarrow$ 0.01 μ F to GND	–	–	–	–	–
Timing Network	Pins 2 and 6 $\rightarrow$ 4700 μ F to GND; Pin 7 $\rightarrow$ 26.6M Ω to +5V; 1k Ω between Pins 6 and 7	Pins 6 and 7 $\rightarrow$ 1M Ω to +5V and 9 μ F to GND	–	–	–	–	–
Trigger/Input	–	Pin 2	Pin 14 (clock from IC 555A Pin 3)	Pin 1 via 470 Ω	Input from PC817	–	Base
Output	Pin 3 $\rightarrow$ CD4017 Pin 14 and IC 555B Pin 2	Pin 3 $\rightarrow$ PC817 and BC549 base	Q0-Q6 outputs	Pins 3 and 4 $\rightarrow$ BT66	Output $\rightarrow$ BC549 base	Driven by BC549	Collector $\rightarrow$ Speaker
Reset/Enable	–	–	Pin 13 $\rightarrow$ GND, Pin 15 $\rightarrow$ Pin 6 (Q7 reset)	–	–	–	–
Other Pins	–	–	Pins 9, 11, 12 $\rightarrow$ Not connected	–	–	–	–